Example Case1
CKD (3 Visits)

This VR presents a case of nephropathy with complex complications. Track the case progression and updates to identify the best management approach.

Visit 1
01/01/2023

Visit 1: 01- Interactions (Q & A)

Type	Q	Ans (entered by user)
Greetings	Hi [Patient name], great to see you.	Hi [user name], Nice to see you too!
General Description	How can I help you feel better today?"	I was referred to the emergency room a few days ago when I suddenly passed blood in my urine; surprisingly, it has now resolved, but I am worried it could be a sign of something more serious. The only unusual thing that had happened was a sore throat a week before the bloody urine.
Eating habits	None	None

Visit 1: 02- Patient profile and medical record

Category	Timestamp	Metric	Value
Previous Measures	Initial (before first visit)	General state	31 years old female normally fit and well.
		Height	NA
		Weight	NA
		BMI	NA
		Blood pressure	NA
		SpO2	NA
		Temperature and other metrics	NA
History and Habits	Initial (before first visit)	**Past history:**	• Gestational diabetes when she had her only child 4 years ago. • Normal term pregnancy without complication, vaginal delivery. • Also known to have migraine.
		Family History	father has hypertension, Mother has type 2 DM.
		Habits	Non-smoke
		Medication History	Ibuprofen PRN for headache

Visit 1: 02- Patient profile and medical record

Category	Timestamp	Metric	Value
Previous Lab Tests	Initial (before the first visit)	None	NA
Previous Scans and Reports	Initial (before the first visit)	None	NA
Current medications and diagnosis:	Initial (before the first visit)	Current illness and diagnosis	None
		Current Medication	None

Visit 1: 03-Physical Examination

Assessment	Evaluation (Possible states)
Pulse and SpO2	75/min
Blood pleasure BP	145/84 mmHg
Body temperature	**Afebrile**
Jugular Venous Pressure (JVP)	• Normal
Skin Colour (Pallor)	• Pink (Normal)
Skin Exposure	• No Abnormality
Edema	• **No Edema (+1 Grade)**
Abdomen **Palpation**	• Normal in all quarters
Lung auscultation	• Clear (Normal)
Heart auscultation	• Normal (S1 or S2)
Height	160 cm
Weight	69 kg
BMI	27

Visit 1: 04-Lab Tests, Scans, and Procedures

Requested Scans.

Their results will be displayed in the table (patient profile) with a timestamp: Visit data + 7 days (07/01/2023)

Timestamp	Type	Report	Pic
07/01/2023	Kidney US	• **Normal size kidneys (Rt kidney 10.1 cm and 10.8 cm for Lt kidney)** • **Increase echogenicity of both kidney** • **No hydronephrosis** • **No stones** • **Normal urinary bladder**	~~NA~~

Visit 1: 04-Lab Tests, Scans, and Procedures

Requested lab tests.

Their results will be displayed in the table (patient profile) with a timestamp: Visit data + 7 days (07/01/2023)

Test	Value
HbA1c	5.9%
Fasting Blood Glucose	72 mg/dL
Pregnancy Test	Negative
iPTH (Intact Parathyroid Hormone)	90 pg/mL
CBC (Complete Blood Count)	- Hemoglobin (Hb): 11.8 gm/dL **- White Blood Cells (WBC)**:6.9 × 10^3/u -Platelets: 246 × 10^3/uL
Serum Electrolytes	-Sodium (Na): 139 mmol/L -**Potassium (K)**: 4.5 mmol/L - Calcium: 2.3 mmol/L
Urine Analysis	-Urine Dipstick: 3+ protein & 3+ RBC (indicates proteinuria and hematuria) **-Urine Microscopy**:40 RBCs & 3 WBCs (indicates hematuria and mild infection)
Kidney Functions	**--Creatinine**:1.7 mg/dL -**Urea**:48 mg/dl -**eGFR (Estimated Glomerular Filtration Rate)**:42 mL/min/1.73m^2 **-UACR (Urine Albumin Creatinine Ratio)**:178 mg/g

Test	Value
Complements C3 & C4	- C3: Normal - C4: Normal
Anti-nuclear antibodies (ANA)	Negative
Anti GBM	Negative
Antineutrophil Cytoplasmic Antibodies (ANCA)	Negative
Antistreptolysin O Test	Normal

Visit 1: 05- CKD classification assessment

-**eGFR (Estimated Glomerular Filtration Rate)**:42 mL/min/1.73m^2
-**UACR (Urine Albumin Creatinine Ratio)**:178 mg/g

KDIGO: Classification and prognosis of CKD				Persistent albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30-300 mg/g 3-30 mg/mmol	>300 mg/g >30 mg/ mmol
eGFR categories (mL/min/ 1.73 m^2) Description and range	G1	Normal or high	≥90	**Low***	**Moderately increased**	**High**
	G2	Mildly decreased	60-89	**Low***	**Moderately increased**	**High**
	G3a	Mildly to moderately decreased	45-59	**Moderately increased**	**High**	**Very high**
	G3b	Moderately to severely decreased	30-44	**High**	**Very high**	**Very high**
	G4	Severely decreased	15-29	**Very high**	**Very high**	**Very high**
	G5	Kidney failure	<15	**Very high**	**Very high**	**Very high**

Visit 1: 06-Pass/Fail stage questions

Question	Choices	Correct answers
Do you think the patient should be hospitalized?	1- true **2- false**	2
Owing to patients' examination and investigations results, patient's main health problem is :	1. DM **2. HTN** **3. Proteinuria**	2 and 3
Owing to patient's condition, management should encompass:	1. DASH diet 2. Avoid NSAIDs, such as ibuprofen 3. Lifestyle modification to control hypertension 4. ACE inhibitor (**5 mg of Ramipril**)	All (1, 2 , 3, and 4)

Visit 1: 06-Pass/Fail stage questions

Question	Choices	Correct answers
Owing to Reem's history and investigations results, Differential diagnosis includes :	1. AKI secondary to NSAID 2. Chronic kidney disease secondary to NSAID 3. AKI on top of CKD 4. Glomerulonephritis	All (1, 2, 3, and 4)
Do you think that this patient has SLE?	1. Yes 2. No	2
SLE can be excluded from differential diagnosis as:	1. Complements are normal 2. Anti-nuclear antibodies are normal 3. Anti GBM is normal	1 and 2
Is there any indication for a kidney biopsy? (needed in the next visit?)	1. Yes 2. No	1

Visit 1: Facilitator Recap in case of success:

Facilitator Recap in case of success:

Finally, remember that:

- Reem recently complained of hematuria.
- Her current medication is Ramipril 5 mg.
- Her lab tests show a Urine Albumin Creatinine ratio of 178 mg/g and an estimated glomerular filtration rate of 42 mL/min/1.73m^2.
- Reem's leading health problems are HTN and proteinuria.
- She has been asked for more investigations.

Visit 2
07/02/2023

Visit 2: 01- Interactions (Q & A)

Type	Q	Ans (entered by user)
Greetings	Hi [Patient name], great to see you.	Hi [user name], Nice to see you too!
General Description	how are you feeling with the new medication?	Actually, doctor, I am feeling better, but my blood pressure has remained the same for the past 10 days;

Visit 2: 02- Patient profile and medical record

Category	Timestamp	Metric	Value
Previous Measures	07/01/2023	Blood pressure The rest of the data in table	145/84 mmHg Visit 1: 03-Physical Examination
Lab tests	07/01/2023	Data in the table =>	Visit 1: 04-Lab Tests, Scans, and Procedures
Scans and reports	07/01/2023	Ultrasound KUB	Visit 1: 04-Lab Tests, Scans, and Procedures
Medication and diagnoses	07/01/2023	Illness and diagnosis	1. HTN 2. Proteinuria Her differential diagnosis includes: 1. AKI secondary to NSAID 2. Chronic kidney disease secondary to NSAID 3. AKI on top of CKD 4. Glomerulonephritis
		Management	1. DASH diet 2. Avoid NSAIDs, such as ibuprofen 3. Lifestyle modification to control hypertension 4. ACE inhibitor (**5 mg of Ramipril**)

Visit 2: 03-Physical Examination

Assessment	Evaluation (Possible states)
Pulse	75 beats/min
Blood pleasure BP	151/93 mmHg
Body temperature	**Afebrile**
Jugular Venous Pressure (JVP)	• Normal
Skin Colour (Pallor) (Visual)	• Pink (Normal)
Skin Exposure (Visual)	• No Abnormality
Edema	• **No Edema (+1 Grade)**
Abdomen **Palpation**	• Normal in all quarters
Lung auscultation (Sound)	• Clear (Normal)
Heart auscultation (Sound)	• Normal (S1 or S2)
BMI	• 27

Visit 2: 04-Lab Tests, Scans, and Procedures

Requested lab tests.

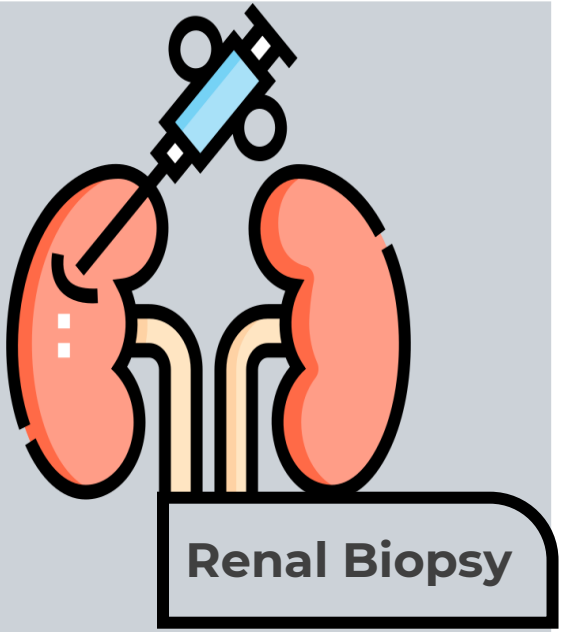
Their results will be displayed in the table (patient profile) with a timestamp: Visit data + 7 days (14/02/2023)

Timestamp	Test (set by the user from outside)	Value set by the user from the form
14/02/2023	Kidney Functions	• **GFR: 43 mL/min/1.73m^2** • **UACR: 160 mg/g**

Visit 2: 04-Lab Tests, Scans, and Procedures

Requested Scans/procedures.

Their results will be displayed in the table (patient profile) with a timestamp: Visit data + 7 days (14/02/2023)

Timestamp	Type	Report	Pic
14/02/2023	**Renal Biopsy**	**Microscopic Description** • Renal cortex and medulla in which up to 15 glomeruli are present in 7 of which are globally sclerosed while another 3 reveal lesions of segmental sclerosis. • There is moderate irregular mesangial expansion due to increase in mesangial matrix as well as cellularity. • There is moderate tubular atrophy and interstitial fibrosis (25-30% of the cortical tissue), associated with moderate multifocal chronic interstitial infiltration by inflammatory cells predominantly lymphocytes with scattered plasma cells. • Small arteries and arterioles reveal moderate medial hypertrophy with focal hyalinosis. Larger arteries are not available for evaluation. **Direct immunofluorescent microscopy** • 8 glomeruli 4 of which are globally sclerosed. • There is glomerular deposition of immunoglobulins (IgG 1+, IgM trace, IgA 2-3+), complement component (C3, 2+), as well as kappa (1+) and lambda (2+) light chains. • The distribution is predominantly within the mesangial areas. There is no deposition of C1q and fibrinogen. • No significant extra glomerular deposition is noted.	

Visit 2: 05- CKD classification assessment

- **GFR: 43 mL/min/1.73m²**
- **UACR: 160 mg/g**

KDIGO: Classification and prognosis of CKD				Persistent albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
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eGFR categories (mL/min/1.73 m²) Description and range	G1	Normal or high	≥90	**Low***	**Moderately increased**	**High**
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	G3b	Moderately to severely decreased	30-44	**High**	**Very high**	**Very high**
	G4	Severely decreased	15-29	**Very high**	**Very high**	**Very high**
	G5	Kidney failure	<15	**Very high**	**Very high**	**Very high**

Visit 2: 06-Pass/Fail stage questions

Question	Choices	Correct answers
What is the likely diagnosis for Reem based on the findings from the kidney biopsy?	1. Membranoproliferative Glomerulonephritis 2. IgA nephritis 3. Lupus nephritis 4. Thin Basement Membrane Nephropathy	2
According KDIGO 2024, Empagliflozin can be used in CKD patients without diabetes, to reduce CKD progression? (show hint 1slide 24 to answer this question)	1. Yes 2. No	1
Empagliflozin safely reduced the risk of kidney disease progression and eGFR decline in all primary kidney disease including IgAN? (show hint 2 slide 25 to answer this question)	1. Yes 2. No	1
According to KDIGO 2024, once Empagliflozin is initiated, it is reasonable to be continued even if the eGFR falls below 20 ml/min/1.73 m2, unless it is not tolerated or KRT is initiated. (show hint 3 slide 26 to answer this question)	1. Yes 2. No	1
What are the monitoring parameters for Reem?	1. Serum electrolytes 2. Kidney function tests 3. HbA1c 4. LDL 5. Renal doppler 6. Plasma Aldosterone renin ratio 7. Thyroid profile	1, 2, 3, and 4
What is the best management plan for the recent diagnosis of IgA nephropathy and high blood pressure reading? (show hint 4 slide 27 to answer this question)	1. Increase Ramipril from 5mg to 10 mg 2. Start on SGLT2 inhibitors 3. A 6-month course of corticosteroid therapy may be considered 4. Life style modification (low salt diet, regular exercise, and healthy diet) 5. Start on Atorvastatin	1, 2,3, 4, and 5

Visit 2: Facilitator Recap in case of success:

Facilitator Recap in case of success:

Finally, remember that:

- Reem was diagnosed as **IgA nephropathy**.
- She was classified **as very high-risk** progression of end stage kidney disease.
- Reem's problem list now includes **hypertension** and **IgA nephropathy**.

Her current management:

- Increase **Ramipril to 10 mg**
- **Start on SGLT2 inhibitors** and **Atorvastatin**
- **Life style modification** (low salt diet, regular exercise, and healthy diet)

Hints to be displayed as a guide for the Questions in Visit 2

KDIGO 2021 guideline for the management of Immunoglobulin A nephropathy

According to KDIGO 2021 guideline for the management of Immunoglobulin A nephropathy (IgAN):

- The primary focus of IgAN management should be multifaceted, optimized supportive care, including:

✓ RAS blockade, as much as tolerated or allowed, blood pressure control, cardiovascular risk minimization, adherence to lifestyle advice, including dietary counseling, smoking cessation, weight control, and exercise as appropriate.

✓ RAS blockade should be instituted irrespective of hypertension if the patient has proteinuria >0.5 g/d (Grade 1B).

- If proteinuria stays above 0.75–1 g/d despite at least 90 days of optimized supportive care:

✓ The patient has a high risk of progressive loss of kidney function and may be considered for a 6-month course of glucocorticoid therapy (Grade 2B).

VO: According to KDIGO 2021 guideline for the management of Immunoglobulin A nephropathy, the patient should continue on Ramipril, SGLTi, life style modification can be considered.

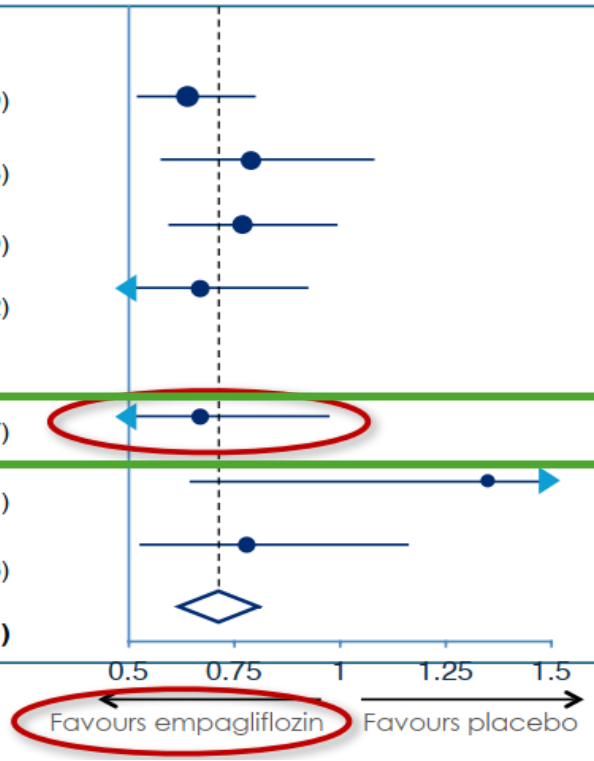
https://kdigo.org/wp-content/uploads/2017/02/KDIGO-2021-GD-Guideline-Executive-Summary.pdf. Last accessed July 2024.

Empagliflozin safely reduced the risk of kidney disease progression and eGFR decline in all primary kidney disease including IgAN.[1]

VO: Empagliflozin safely reduced the risk of kidney disease progression and eGFR decline in all primary kidney disease including IgAN.

Kidney disease progression by primary kidney disease aetiology including types of glomerular disease

Subgroup	Empagliflozin Events/participants (n/n)	Placebo Events/participants (n/n)	Hazard ratio (95% CI)	Heterogeneity test
Causes of kidney disease				
Diabetic kidney disease	137/1032	189/1025	0.64 (0.52, 0.80)	*p*=0.62
Hypertensive/ renovascular disease	72/706	87/739	0.79 (0.58, 1.08)	
Glomerular disease	115/853	139/816	0.77 (0.60, 0.99)	
Other disease/unknown	60/713	89/725	0.67 (0.48, 0.92)	
Types of glomerular disease				
IgAN	51/413	67/404	0.67 (0.46, 0.97)	*p*=0.25
Focal segmental glomerulosclerosis	17/98	13/97	1.35 (0.65, 2.81)	
Other glomerulonephritis	47/342	59/315	0.78 (0.53, 1.16)	
All participants			**0.71 (0.62, 0.81)**	



IgAN: IgA nephropathy

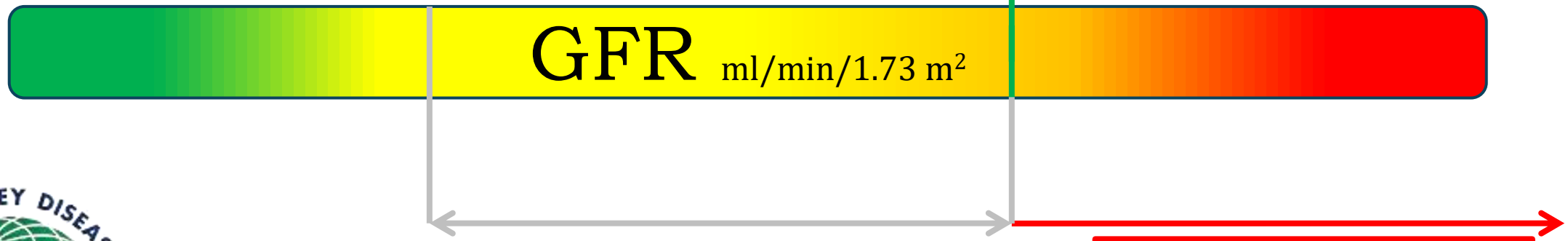
Judge, P. K., Staplin, N., Mayne, K. J., Wanner, C., Green, J. B., Hauske, S. J., ... & Becker, L. (2024). Impact of primary kidney disease on the effects of empagliflozin in patients with chronic kidney disease: secondary analyses of the EMPA-KIDNEY trial. The Lancet Diabetes & Endocrinology, 12(1), 51-60.

2024 KDIGO

VO: The KDIGO 2024 guidelines suggested treatment with an SGLT2 inhibitor for adults with an eGFR of 20 to 45 ml/min/1.73 m² and a urine ACR < 200 mg/g

SGLT2 inhibitors are recommended if urine ACR ≥ 200 mg/g

Adults with CKD with an eGFR ≥ 20*



Adults with eGFR 20* to 45

SGLT2 inhibitors are suggested

Initiation not recommended with eGFR <20*.[2]

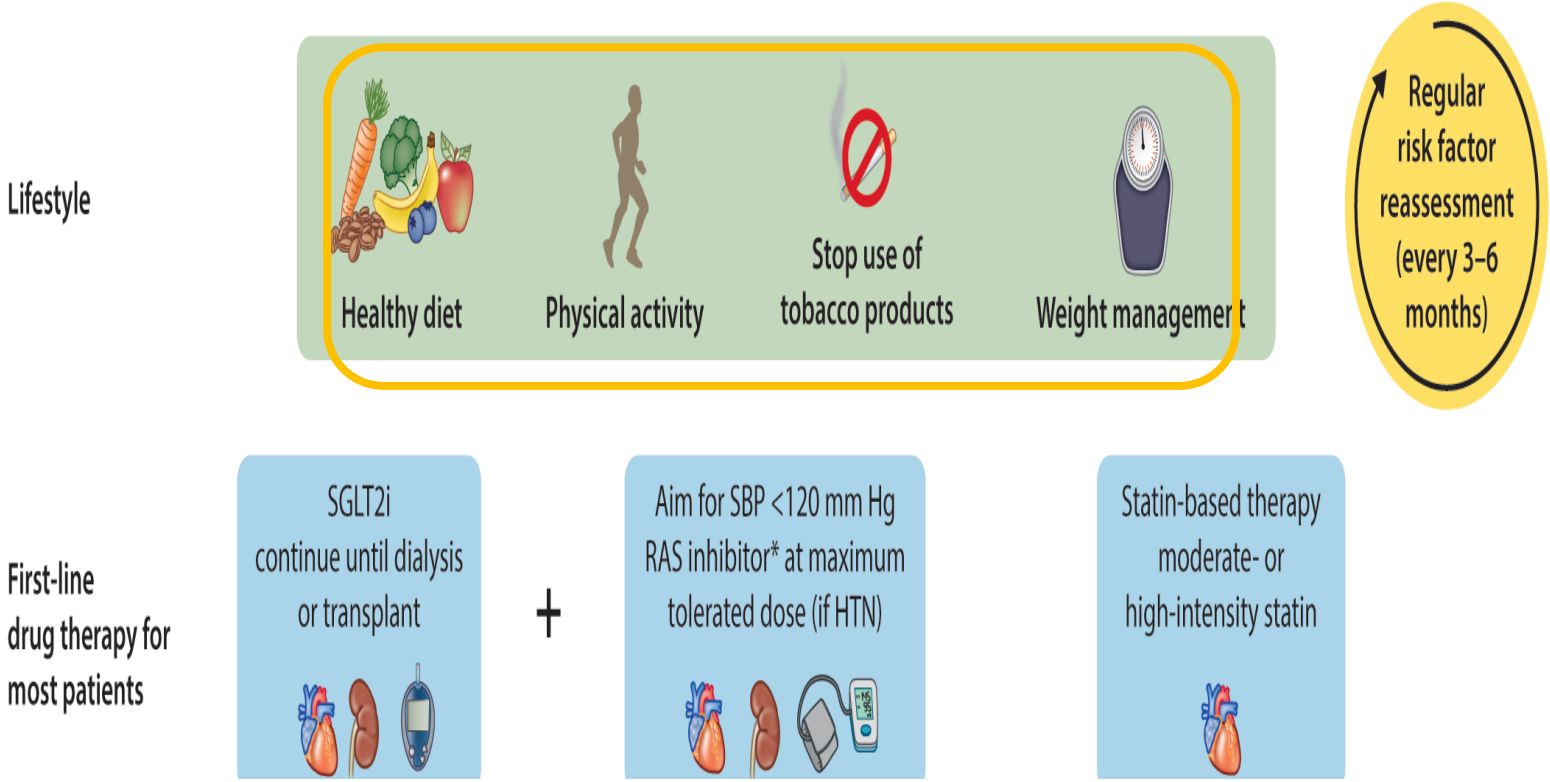
* eGFR ≥ 25 for Dapagliflozin

Practice Point 3.7.1: Once an SGLT2i is initiated, it is reasonable to continue an SGLT2i even if the eGFR falls below 20 ml/min per 1.73 m², unless it is not tolerated or KRT is initiated.

1. Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group (2024). KDIGO 2024 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney international, 105(4S), S117–S314. https://doi.org/10.1016/j.kint.2023.10.018
2. de Boer, I. H., Khunti, K., Sadusky, T., Tuttle, K. R., Neumiller, J. J., Rhee, C. M., Rosas, S. E., Rossing, P., & Bakris, G. (2022). Diabetes Management in Chronic Kidney Disease: A Consensus Report by the American Diabetes Association (ADA) and Kidney Disease: Improving Global Outcomes (KDIGO). Diabetes care, 45(12), 3075–3090. https://doi.org/10.2337/dci22-0027

Management – Comprehensive Treatment Strategy

VO: The KDIGO guidelines devised an algorithm to guide physicians on how to treat patients with CKD. This includes starting on lifestyle modifications all through managing complications that could be associated with CKD.



Treat people with CKD with a comprehensive treatment strategy to reduce risks of progression of CKD and its associated complications encompassing education, lifestyle, exercise, smoking cessation, diet, and medications, where indicated.

SGLT2 : Sodium-Glucose Transport Protein 2, SBP: systolic blood pressure, RAS: Renin Angiotensin system, GLP-1: Glucagon-like peptide-1, MRA: Mineralocorticoid receptor antagonist CCB: calcium channel blocker, ASCVD: Atherosclerotic cardiovascular disease, CKD-MBD: Chronic kidney disease-mineral and bone disorder, PCSK9: proprotein convertase subtilisin/kexin type 9, CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate

KDIGO 2024

Visit 3 (after 8 weeks)
14/04/2023

Visit 3: 01- Interactions (Q & A)

Type	Q	Ans (entered by user)
Greetings	Hi [Patient name], great to see you.	Hi [user name], Nice to see you too!
General Description	how are you feeling with the new medication?	I am fine doctor, I have been doing well and have been tolerating my treatment

Visit 3: 02- Patient profile and medical record

Category	Timestamp	Metric	Value
Previous Measures	07/01/2023	Blood pressure The rest of the data in table	145/84 mmHg Visit 1: 03-Physical Examination
	07/02/2023	Blood pressure	151/93 mmHg Visit 2: 03-Physical Examination
Lab tests	07/01/2023	Data in the table =>	Visit 1: 04-Lab Tests, Scans, and Procedures
	14/02/2023	Data in the table =>	Visit 2: 04-Lab Tests, Scans, and Procedures
Scans and reports	07/01/2023	Ultrasound KUB	Visit 1: 04-Lab Tests, Scans, and Procedures
	14/02/2023	Renal Biopsy	Visit 2: 04-Lab Tests, Scans, and Procedures
Medication and diagnoses	07/01/2023	Illness and diagnosis	1. HTN 2. Proteinuria Her differential diagnosis includes: 1. AKI secondary to NSAID 2. Chronic kidney disease secondary to NSAID 3. AKI on top of CKD 4. Glomerulonephritis
		Management	1. DASH diet 2. Avoid NSAIDs, such as ibuprofen 3. Lifestyle modification to control hypertension 4. ACE inhibitor (**5 mg of Ramipril**)
	14/02/2023	Illness and diagnosis	**hypertension** and **IgA nephropathy**
		Management	**1. Increase Ramipril to 10 mg** **2. SGLT2 inhibitors** and **Atorvastatin** **3. Lifestyle modification** (low salt diet, regular exercise, and healthy diet)

Visit 3: 03-Physical Examination

Assessment	Evaluation (Possible states)
Pulse	75 beats/min
BMI	26
Blood pleasure BP	110/70 mmHg
Body temperature	**Afebrile**
Jugular Venous Pressure (JVP)	• Normal
Skin Colour (Pallor) (Visual)	• Pink (Normal)
Skin Exposure	• No Abnormality
Edema	• **No Edema (+1 Grade)**
Abdomen **Palpation**	• Normal in all quarters
Lung auscultation	• Clear (Normal)
Heart auscultation	• Normal (S1 or S2)

Visit 3: 04-Lab Tests, Scans, and Procedures

Requested lab tests.

Their results will be displayed in the table (patient profile) with a timestamp: Visit data + 0 days (the patient had them with him when he came for the visit)

Timestamp	Test (set by the user from outside)	Value set by the user from the form
14/04/2023	HbA1c	5.4%
	Lipid profile	LDL: 100 mg/dL
	Serum Electrolytes	-Potassium (K): 4.2 mmol/l Calcium: 2.6 mmol/L
	Kidney Functions	-**eGFR**:51 mL/min/1.73m^2 **- UACR:** 61 mg/g

Visit 3: 05- CKD classification assessment

- **GFR: 51 mL/min/1.73m²**
- **UACR: 61 mg/g**

KDIGO: Classification and prognosis of CKD				Persistent albuminuria categories Description and range		
				A1	A2	A3
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eGFR categories (mL/min/ 1.73 m²) Description and range	G1	Normal or high	≥90	**Low***	**Moderately increased**	**High**
	G2	Mildly decreased	60-89	**Low***	**Moderately increased**	**High**
	G3a	Mildly to moderately decreased	45-59	**Moderately increased**	**High**	**Very high**
	G3b	Moderately to severely decreased	30-44	**High**	**Very high**	**Very high**
	G4	Severely decreased	15-29	**Very high**	**Very high**	**Very high**
	G5	Kidney failure	<15	**Very high**	**Very high**	**Very high**

Visit 3: 06-Pass/Fail stage questions

Question	Choices	Correct answers
Do you suggest the patient to continue on the same medication plan?	1. Yes 2. No	1
What are the investigations needed for Reem for the next follow up?	1. Kidney function tests 2. HbA1c 3. LDL 4. Renal doppler 5. Plasma Aldosterone renin ratio 6. Thyroid profile	1, 2, and 3,

Visit 3: Facilitator Recap in case of success:

Facilitator Recap in case of success:

Congratulations on reaching the final stage! You've completed the challenge. It's been a pleasure guiding you through it. I hope this was a valuable learning experience. Keep exploring and expanding your knowledge.

Finally, remember that:

- Reem was diagnosed as **IgA nephropathy**.
- She was classified **as high-risk** progression of end stage kidney disease.
- Reem's problem list now includes **hypertension** and **IgA nephropathy**.

Her current management:

- Continue on **Ramipril 10 mg**
- **SGLT2 inhibitors** and **Atorvastatin**
- **Life style modification** (low salt diet, regular exercise, and healthy diet)

After 6 Months of the current management:

- **She is classified as Moderately increased** progression of end-stage kidney disease with **(eGFR**:62 mL/min/1.73m^2, and UACR: 33 mg/g)
- Her BP is stable at 108/77 mmHg, BMI is 23 kg/m^2, and HbA1c is 4.9%

Finally, remember that **Key take aways**:

- Refer hematuria patient to nephrologist, as there may be an underlying chronic kidney disease
- Criteria for diagnosis and risk stratification of CKD require both eGFR and UACR
- Empagliflozin can be used to manage patients with chronic kidney disease with or without T2D across their disease journey